

C.2.2. Illustration of Reasoning Step Generalization

Reasoning step generalization refers to the scenario where the number of transformations in the test compositional transformation is different from that in the training compositional transformation.

```
Training:
A P P L [F2] [F2] <think> P P L A [F2] <answer> P L A P
Test:
A P P L [F2] <think>
Test:
A P P L [F2] [F2] [F2] <think>
```

C.3. Illustration of Format Generalization

Format generalization refers to scenarios where the input formats are changed during testing. In this work, we consider three different mechanisms that the format gets changed: addition, deletion and modification: **Insert**. It refers to an additional unknown token that is added to the task. The following example shows how the APPL gets changed under addition:

```
Training:
A P P L [F2] [F2] <think> P P L A [F2] <answer> P L A P
Test:
A P <noise> P L [F2] [F2] <think>
```

Delete refers to tokens that are deleted during the test. The following example shows how the APPL gets changed under delete:

```
Training:
A P P L [F2] [F2] <think> P P L A [F2] <answer> P L A P
Test:
A P L [F2] [F2] <think>
```

Modify. It refers to tokens that are replaced by an unknown token during the test. The following example shows how the APPL gets changed under modification:

```
Training:
A P P L [F2] [F2] <think> P P L A [F2] <answer> P L A P
Test:
A <noise> P L [F2] [F2] <think>
```

D. Experiment Environment and Implementation Details

D.1. Environment Setup

We conduct controlled experiments using LLMs with different sizes (ranging from 62K to 3B) and different architectures (GPT and LLaMA) of LLMs; detailed hyperparameters are summarized in Table 3. For LLMs trained from scratch, we employ the AdamW optimiser in mixed precision (FP16). The default learning rate is 3×10^{-3} , and the schedule follows a cosine decay with a 10% warm-up ratio. Training is conducted using a batch size of 1024, and each model is optimized for 10 epochs. A

weight decay of 0.01 is applied, and gradient norms are clipped at 1.0. During the inference time, we set the temperature to 1e-5.

For fine-tuning state-of-the-art LLMs, we use a per-device batch size of 16 with 8 gradient accumulation steps (effective batch size of 128) and train for 24K optimization steps. We set the learning rate to 1e-4 and adopt a cosine learning-rate schedule with a 10% warm-up ratio. All fine-tuning experiments are conducted with bfloat16 (bf16) mixed-precision training. Given the scale of the data, the results of controlled experiments are averaged over three independent runs.

Table 3 | Hyperparameter settings for LLMs with different model sizes and architectures

Architecture	# Params	Hidden Size	Intermediate Size	# Layers	# Heads
GPT	68K	32	N/A	4	4
	589K	80	N/A	7	8
	4.8M	256	N/A	6	4
	35M	512	N/A	11	8
	540M	1536	N/A	19	24
	3B	3072	N/A	26	32
LLaMA	62K	48	128	2	4
	631K	80	216	8	4
	6M	288	768	6	6
	60M	640	1728	12	10
	623M	1536	4096	22	12
	3B	3072	8192	26	24

D.2. Computational Cost

We conduct training, fine-tuning, and inference on utilizing 8 NVIDIA A100 GPUs (80 GB memory) and 4 NVIDIA H200 GPUs.

E. Additional Theory and Propositions

E.1. Task Distribution

To investigate the extent to which CoT reasoning can handle *tasks* under various distribution discrepancies, we design task generalization experiments. As we discussed in Section 4, we decompose tasks into a combination of various *transformations* and *elements*. Therefore, we consider task generalization from two dimensions: transformation generalization and element generalization.

Task Generalization Complexity. Guided by the data distribution lens, we first introduce a measure for generalization difficulty:

Proposition E.1 (Task Generalization Complexity). *For a reasoning chain f_s operating on elements $\mathbf{e} = (a_0, \dots, a_{l-1})$, define:*

$$\text{TGC}(C) = \alpha \sum_{i=1}^m \mathbb{1} [a_i \notin \mathcal{E}_{\text{train}}^i] + \beta \sum_{j=1}^n \mathbb{1} [f_j \notin \mathcal{F}_{\text{train}}] + \gamma \mathbb{1} [(f_1, f_2, \dots, f_k) \notin \mathcal{P}_{\text{train}}] + C_T \quad (14)$$

as a measurement of task discrepancy Δ_{task} , where α, β, γ are weighting parameters for different novelty types and C_T is task specific constant. $\mathcal{E}_{\text{train}}^i, \mathcal{F}_{\text{train}}$, and $\mathcal{P}_{\text{train}}$ denote the bit-wise element set, relation set, and the order of relation set used during training.

We establish a critical threshold beyond which CoT reasoning fails exponentially:

Theorem E.1 (Task Generalization Failure Threshold). *There exists a threshold τ such that when $TGC(C) > \tau$, the probability of correct CoT reasoning drops exponentially:*

$$P(C) \leq e^{-\delta(TGC(C)-\tau)} \quad (15)$$

The proof is provided in Appendix H.2.

E.2. Length Distribution

Length generalization examines how CoT reasoning degrades when models encounter test cases that differ in length from their training distribution. The difference in length could be introduced from the text space or the reasoning space of the problem. Therefore, we decompose length generalization into two complementary aspects: text length generalization and reasoning step generalization. Guided by instinct, we first propose to measure the length discrepancy.

Proposition E.2 (Length Extrapolation Gaussian Degradation). *For a model trained on chain-of-thought sequences of fixed length L_{train} , the generalization error at test length L follows a Gaussian distribution:*

$$\mathcal{E}(L) = \mathcal{E}_0 + (1 - \mathcal{E}_0) \cdot \left(1 - \exp \left(-\frac{(L - L_{train})^2}{2\sigma^2} \right) \right) \quad (16)$$

where $\mathcal{E}_0$ is the in-distribution error at $L = L_{train}$, σ is the length generalization width parameter, and L is the test sequence length

The proof is provided in Appendix H.3.

E.3. Format Distribution

Format generalization assesses the robustness of CoT reasoning to surface-level variations in test queries. This dimension is especially crucial for determining whether models have internalized flexible, transferable reasoning strategies or remain reliant on the specific templates and phrasings encountered during training.

Format Alignment Score. We introduce a metric for measuring prompt similarity:

Definition E.1 (Format Alignment Score). *For training prompt distribution P_{train} and test prompt p_{test} :*

$$PAS(p_{test}) = \max_{p \in P_{train}} \cos(\phi(p), \phi(p_{test})) \quad (17)$$

where ϕ is a prompt embedding function.

Table 4 | Evaluation on transformation generalization.

Transformation (Train → Test)	Exact Match (%)			Edit Distance			BLEU Score		
	Reasoning	Answer	Full Chain	Full Chain	Reasoning	Answer	Full Chain	Reasoning	Answer
$\{f_2 \circ f_3, f_3 \circ f_2, f_3 \circ f_3\} \rightarrow f_2 \circ f_2$	6.66	10.25	6.66	0.0941	0.0718	0.2244	0.5417	0.6683	0.1982
$\{f_2 \circ f_3, f_3 \circ f_2, f_2 \circ f_2\} \rightarrow f_3 \circ f_3$	100.00	9.19	9.19	0.0488	0.0000	0.1768	0.8220	1.0000	0.1932
$f_2 \circ f_3 \rightarrow f_3 \circ f_2$	0.00	0.00	0.00	0.2997	0.3728	0.4808	0.2000	0.0019	0.0000
$f_3 \circ f_2 \rightarrow f_2 \circ f_3$	0.00	0.00	0.00	0.2334	0.2249	0.4808	0.2548	0.0952	0.0000